

Provisioning, Connection, and the Unfinished Loop

A Thesis Report on Recursive Ecosystem Formation in Tamil Nadu's State-Led Startup Programs

*Companion analysis to the essay series · Evidence current to mid-2026 · All load-bearing claims graded **Confirmed** or **Emerging**, with causal claims separately flagged, and referenced with source links.*

Abstract

A startup ecosystem is not a stockpile of inputs but a set of self-reinforcing feedback loops in which outputs re-enter as inputs — founders producing employees who become founders, exits producing capital that funds new ventures. This thesis asks whether Tamil Nadu's state-led startup programmes are producing such recursion, or merely provisioning and attempting to connect components. Using a Component–Connection–Closure framework and a primary-source verification protocol, it examines nine initiatives through seven structured case studies.

It finds that provisioning is comprehensive, connection is deliberately engineered, and recursion is genuinely evidenced in several places. But two findings qualify any celebratory reading. **First, on causation:** the two fully closed loops (the Zoho/Freshworks alumni network and IIT Madras Research Park) were inherited rather than state-created, and the flagship instrument, TANSEED, cannot be shown to *cause* the outcomes it is associated with. Because TANSEED selects already-promising startups, its association with roughly ₹535 crore of downstream private capital is as consistent with skilled selection as with catalysis — and the celebrated cases are precisely those where selection dominates. The state's *evidenced* role is therefore credentialer, convener, and selector, not originator. **Second, on distribution:** every measured loop closes through follow-on funding, which dilutes founders and channels equity to external investors. Recursion and ownership-broadening are distinct axes, and the one fully closed loop is narrow and elite (urban, English-educated SaaS), while the instruments designed to broaden ownership — the SC/ST fund, the women-led track, rural and tier-2 programmes — are exactly the loops that have not closed.

The thesis concludes that ecosystem formation in Tamil Nadu is real and accelerating, but it is neither clearly state-caused nor, as yet, ownership-widening. Recursion is necessary but not sufficient; closure among the already-advantaged is not the same as broad-based ownership.

Chapter 1 — Introduction

1.1 Background

Over the past decade Tamil Nadu has moved from a predominantly services-and-manufacturing economy toward an explicit innovation agenda, institutionalised through the Tamil Nadu Startup and Innovation

Mission (StartupTN/TANSIM, established 2018) and successive startup policies [31], [30]. The state has registered thousands of startups, issued tens of thousands of “Smart Cards,” and stood up a portfolio of capital, talent, market-access, and deep-tech instruments [7].

1.2 Problem Statement

Policy evaluations of startup ecosystems frequently count *components* — grants disbursed, startups registered, events held — and infer health from their volume. But components are necessary, not sufficient. The entrepreneurial-ecosystem literature treats ecosystems as systems defined by feedback, not inventory [1]. This thesis addresses two gaps that volume metrics conceal: the gap between *provisioning activity* and *recursive self-sustenance*, and — once recursion is found — the gap between *correlation* and *causation*, and between *ecosystem formation* and *ownership distribution*.

1.3 Research Questions

- **RQ1.** Which components has Tamil Nadu provisioned, and across how much of the startup stack?
- **RQ2.** What deliberate mechanisms connect these components?
- **RQ3.** For which initiatives is recursion (loop closure) evidenced, and for which is it only attempted?
- **RQ4.** What distinguishes loops that have closed from those that have not?
- **RQ5.** Is Tamil Nadu’s startup ecosystem becoming self-sustaining, or does it remain state-dependent?
- **RQ6.** Where recursion is evidenced, is there evidence the state *caused* it — or selected startups that would likely have succeeded regardless?
- **RQ7.** Does loop closure broaden ownership, or can a recursing ecosystem concentrate it?

1.4 Objectives

To classify initiatives along the provisioning–connection–closure chain; to verify load-bearing claims against primary sources while distinguishing relationship from causation; to identify the structural conditions associated with closure; to test whether closure broadens or concentrates ownership; and to propose falsifiable indicators by which future self-sustenance can be judged.

1.5 Scope and Significance

The study covers nine StartupTN-adjacent initiatives and two organic/institutional anchors, using public records to mid-2026. It is significant for its transferable method (Component–Connection–Closure, with explicit causal and distributional tests) and for framing its conclusions as testable predictions rather than advocacy.

Chapter 2 — Conceptual Framework and Literature

2.1 Ecosystems as recursive systems

An entrepreneurial ecosystem becomes self-sustaining when outputs feed back as inputs. The strongest form is the **founder flywheel**: a successful company trains employees, some found new companies, some

succeed and then *invest in or mentor* the next generation. Indian venture literature documents this in the “mafia” pattern — clusters spun out of a single parent, with founders recirculating as angels and mentors [1], [2].

2.2 The Component–Connection–Closure framework

Three terms operationalise recursion:

- **Component** — a provisioned input (a grant, a hackathon, a research park).
- **Connection** — a deliberate mechanism linking components.
- **Closure** — evidence that a connection produces a feedback effect (a funded founder who later mentors or finances successors).

The benchmark closed loop is: **founder → employees → new founders → investors → new founders**. For each initiative the analysis asks *which turns of this loop are evidenced*. A loop that demonstrates “grant → scale” but not “founder → investor → next founder” is *partially closed*.

2.3 Two axes the framework must keep separate

Recursion is **necessary but not sufficient** for a healthy ecosystem, and it is a *different axis* from ownership distribution. A loop can close vigorously while concentrating ownership, because the usual closure mechanism — follow-on funding — dilutes founders and transfers equity to external investors and eventual acquirers. This thesis therefore treats two questions as distinct:

- **Recursion (formation):** do outputs re-enter as inputs?
- **Ownership (distribution):** who ends up holding the equity and economic control the ecosystem creates — founders and local/community stakeholders, or external capital?

“Ownership-broadening” here means widening the base of people and places that hold durable equity and control, not merely increasing the number of ventures or the volume of capital cycling through them. Conflating the two — treating a closing loop as evidence of widening ownership — is the central analytical hazard this thesis guards against.

2.4 The causal-inference problem

Programme funders select. A seed fund that backs already-promising ventures will show strong downstream outcomes whether or not it *caused* them — the classic selection-versus-treatment problem. Establishing causation requires a counterfactual (a comparison group, a discontinuity around the eligibility threshold, or at minimum a credible mechanism). Co-occurrence (“X received a grant and later raised capital”), even when asserted by the funder, is not a counterfactual. This thesis grades relationship and causation separately.

2.5 Comparators

Internally, the Zoho/Freshworks network and IIT Madras Research Park prove recursion is achievable in Tamil Nadu [2], [4]. Externally, Rajasthan’s iStart procurement model is a benchmark for a *state-built* market-access loop that has closed at scale [24].

Chapter 3 — Methodology

3.1 Research design

A qualitative, multiple-case design. Initiatives are the units of analysis; each is examined on a fixed frame (context, provisioning, connection, recursion evidence, loop status, verification, assessment) to enable cross-case comparison.

3.2 Data sources

Official StartupTN communications; implementer materials (Villgro, Headstart); regulatory and corporate records; the full text of cited third-party reports (notably the Primus Partners procurement report); and named secondary reporting (Inc42, YourStory, Entrackr, Business Standard, The Statesman). All sources are listed with links in **References**.

3.3 Evidence grading

- **Confirmed** — supported by a primary source or by multiple named secondary sources.
- **Emerging** — component and connection documented, but loop closure not yet evidenced.

A separate **causal grade** applies to any claim that a programme *produced* an outcome: **Established** (counterfactual or strong independent mechanism), **Asserted** (claimed, often by the funder, without counterfactual), or **Not tested**.

3.4 Verification protocol

Three rules, adopted after errors in the research process:

1. **Verify the connection, not just the entity.** A real company plus a real funding event does not establish that the funding was *caused by* a programme; each causal link was checked against a source asserting it directly.
2. **Absence of evidence is not evidence of absence.** Unverifiable claims were marked Emerging, not false; several were later Confirmed once primary sources were consulted.
3. **Relationship is not causation.** Confirming that a startup is a programme beneficiary, even via the programme's own communications, confirms the relationship and the funder's claim — not the counterfactual. Funder-sourced causal claims are graded **Asserted**, not Established.

3.5 Limitations

There is **no counterfactual** in this study: no comparison group of similar non-funded startups, and no discontinuity analysis around eligibility thresholds. Causal claims are therefore graded but not proven. Aggregate figures (e.g., follow-on totals) are largely self-reported by StartupTN and unaudited here. A substantial share of beneficiary-level causal claims rests on StartupTN's own posts — the party with the strongest incentive to claim credit. Several programmes are recent (2024–2026), so “Emerging” frequently denotes youth; one exception is noted (TANSEED, §6.2). Distributional (ownership) data at founder/cap-table level is largely unavailable in public records, so §6.3's distributional analysis is directional.

Chapter 4 — The Provisioning and Connection Layers

4.1 Provisioning (RQ1)

Tamil Nadu has an instrument for nearly every layer of the stack:

- **Talent:** Naan Mudhalvan and the Niral Thiruvizha hackathon, reaching ~28,000 students across ~290 institutions in a recent edition [\[27\]](#).
- **Early capital:** TANSEED grants (₹10–15 lakh); a dedicated SC/ST Startup Fund; a women-led track; angel-network and investor-connect programmes; thematic funds [\[7\]](#), [\[8\]](#).
- **Market access:** a procurement framework relaxing eligibility barriers, with direct purchase up to ₹50 lakh via TANSIM [\[24\]](#), [\[31\]](#).
- **Deep-tech anchoring:** IIT Madras Research Park and the iTNT Hub [\[6\]](#), [\[28\]](#).
- **Convening:** the annual Uimage TN summit [\[29\]](#).

RQ1 is answered affirmatively: provisioning is comprehensive and spans the stack.

4.2 Connection (RQ2)

Tamil Nadu's design wires components together:

- **TANSEED** takes a ~3% support stake (from edition five) and enrolls recipients in a year-long accelerator with mentorship and delegations [\[8\]](#).
- **Niral Thiruvizha** issues finalists Smart Cards routing them into hubs and incubators [\[27\]](#).
- **The SC/ST Fund** invests via equity/CCDs, making the state a shareholder [\[22\]](#).
- **iTNT Hub** connects 570+ colleges, researchers, and industry [\[28\]](#).
- **TN-RISE** is a women founders' council and co-creation centre with mentorship MoUs [\[20\]](#).

RQ2 is answered: connection is intentional and architecturally explicit.

Chapter 5 — Findings: Case Studies

Each case answers RQ3 (is recursion evidenced?) and, where relevant, RQ6 (caused or selected?).

5.1 The Zoho / Freshworks Founder Flywheel (benchmark)

Context. Zoho and its spin-out Freshworks made Chennai a leading SaaS hub with no decisive state role.

Provisioning/connection. Organic and cultural: a skills-first employer training engineers who later founded companies. **Recursion evidence.** ~59 former Zoho employees became founders (Freshworks, Chargebee, Facilio, etc.); alumni founders became angels and mentors [\[2\]](#), [\[3\]](#). **Loop status. Closed** — every turn is evidenced. **Verification / causal grade.** Confirmed; not state-caused (organic) [\[1\]](#), [\[2\]](#). **Assessment.** Proof recursion is achievable here — and the reference standard. But the loop is narrow (urban, English-educated SaaS) and emerged organically over ~two decades, not via policy. It is the clearest case where recursion is real *and* ownership remains concentrated among the already-advantaged.

5.2 IIT Madras Research Park / IITMIC (institutional anchor)

Context. India's first university research park plus its incubation cell — the country's leading deep-tech incubator. **Provisioning/connection.** Lab access, faculty linkage, IP support, investor network, and an incubation model that *retains a stake* in companies so founders stay engaged [5]. **Recursion evidence.** 457 startups over ~12 years; survival ~80% (vs typical 8–10%); ~60% non-IIT founders; two unicorns; an IPO (Ather) [4], [6]. **Loop status. Closed (institutional).** Cap-table retention is itself a designed recursion mechanism. **Verification / causal grade.** Confirmed; caused institutionally (not by StartupTN) [4], [5]. **Assessment.** A model for how an *institution* (not a market, and not the state) can manufacture recursion. Inherited by the broader ecosystem.

5.3 TANSEED (the state's flagship — relationship confirmed, causation not established)

Context. Launched 2021; StartupTN's flagship early-capital instrument and the central test of whether the *state* can build a loop. **Provisioning/connection.** ₹10–15 lakh grants; ~3% support stake; year-long accelerator; routing into incubators and investors; ~169 startups supported since 2021 [8]. **Recursion evidence (aggregate).** Against ~₹40 crore allocated, supported startups have collectively raised ~₹535 crore in external follow-on funding; named beneficiaries include Buyofuel, Ippopay, Gallabox, Dream Aerospace, Torus Robotics, Plugzmart, Greenpod [7].

The causal caveat (central to this case). This evidence is co-occurrence, not a counterfactual. TANSEED *selects* promising startups; its association with ₹535 crore of follow-on is as consistent with skilled selection as with catalysis. Crucially, the celebrated cases are precisely those where selection dominates:

- **ippopay** (Mohan K, Chennai; payments). 2021 TANSEED winner; founder calls it the company's "first recognition," after which it scaled. *Turns evidenced: grant → validation → scale. Causal grade: Asserted.* Confirmed relationship [9], [10].
- **PrimeInvestor** (Srikanth Meenakshi, Vidya Bala, Bhavana Acharya; wealthtech). A ₹10 lakh TANSEED 3.0 grant in 2022; a ₹19.5 crore Rainmatter (Zerodha) round in 2026 — its first *external* round. The founders are veteran ex-FundsIndia operators; a four-year-old non-dilutive grant is unlikely to have *caused* a Zerodha-backed round. This is the clearest illustration that beneficiary status and later success can be linked by the funder without a causal mechanism. *Causal grade: Asserted (weak).* Relationship confirmed [11], [12], [13], [14].
- **REUDE Technologies** (Reginald Chellam; drone/UAV). TANSEED 4.0 grantee and the *sole TN finalist* among the top ten at Stanford's Seed Spark — i.e., selected for unusual promise. *Causal grade: Asserted.* Confirmed relationship [15], [16].
- **HARVTECH** (agritech/EV-tractors). Selected from 1,000+ applicants for a TANSEED 4.0 grant; stacked with IIT-M Carbon Zero recognition and a BIRAC BIG grant — again, selected for promise. *Causal grade: Asserted.* Confirmed relationship [17], [18].

The cases that would *demonstrate* causation are the marginal ventures that would not have reached prototype or pilot without ₹10–15 lakh and an accelerator — and those are exactly the ones not celebrated, and for which no data is published.

Loop status. Selection vs treatment unresolved. The beneficiary relationships and the scaling are Confirmed; that TANSEED *caused* the scaling is only Asserted. The grant → validation → scale arc is real; whether the state catalysed it or merely identified it is undetermined — and the strongest evidence skews toward identification. **Assessment.** TANSEED demonstrates that the state is a capable *selector and*

credentialer whose imprimatur co-occurs with private capital. That is a valuable ecosystem function — it lowers VC search costs and confers legitimacy. It is not the same as originating the outcome, and the catalytic claim remains unproven.

5.4 Public Procurement: Tamil Nadu vs. Rajasthan (comparative case)

Context. “Government as first customer” is a textbook recursion mechanism. Tamil Nadu has the policy; a peer state shows the outcomes. **Provisioning/connection.** TN relaxes eligibility barriers and permits direct procurement up to ₹50 lakh [24]. **Recursion evidence (thin).** The Primus Partners report (Table 6) records Tamil Nadu at ~5 startups / ~₹0.54 crore in work orders — near Karnataka (₹0.49 crore) and Telangana (~₹1 crore) [24]. **Contrast — Rajasthan.** Via iStart and e-Bazaar (plus procurement-law reform), Rajasthan issued 132 work orders worth over ₹25 crore to 476 startups; recipient surveys report majorities citing improved market access and ~70% citing increased investor interest after government contracts [24], [25]. Kerala (~141 startups) is a second working model. **Loop status. Open in Tamil Nadu; closed in Rajasthan.** **Verification / causal grade.** Confirmed figures; the Rajasthan loop has at least survey-level mechanism evidence (Established-leaning) [24]. **Assessment.** The most actionable gap, and an instructive contrast: this is the one case where a *state* can plausibly claim to have caused the loop (Rajasthan’s platform + legal reform preceded the orders) — and Tamil Nadu has not built it.

5.5 SC/ST Startup Fund (an ownership-broadening instrument — still open)

Context. An exclusive equity fund for SC/ST founders; an explicit *distributional* instrument. **Provisioning/connection.** ~₹60.8 crore across ~26+ startups (₹7.5 crore to five in batch one; ₹9.75 crore to eight in batch two; ₹34.6 crore to 26 by end-2023), via equity/CCDs with post-investment mentorship [22], [23]. **Recursion evidence.** Named, funded companies — Ecopmin Technologies (₹3 crore), Adhiban Farms (₹2.2 crore), COEO Wellness (₹60 lakh), Mannvasanai Traditional Food (₹45 lakh), Nanjil Nadu Kani Tribal Women Creations (₹25 lakh), TAMS Tribal Green Fuel [22], [23]. **Loop status. Emerging.** Capital and credibility flow to founders; the aim of alumni becoming mentors/role models is not yet documented. **Assessment.** Precisely the kind of instrument the ownership question cares about — and it has not closed. The state’s distributional bets remain at the funding stage, not yet recursive.

5.6 Women-Led TANSEED and TN-RISE (an ownership-broadening instrument — still open)

Context. A women-led TANSEED track plus TN-RISE, a women founders’ council launched July 2024 (World Bank-supported) [20], [21]. **Provisioning/connection.** ₹15 lakh women-led TANSEED awards; TN-RISE as a co-creation centre with mentorship MoUs (Flipkart, HP, FLO) and a stated target of ~1,000+ women mentored per year [21]. (Some StartupTN materials cite larger targets; this thesis uses the conservative figure.) **Recursion evidence.** At the Global Women Summit 2026 (TNWeSafe), ₹15 lakh TANSEED awards went to women-led ventures including Dukil Medical and Technical Textiles (Santhiya Selvakumar, Erode), Aruvi Eco (Kalaiselvi Balamurugan, Tiruppur), and others across Coimbatore, Chennai, and Tiruvannamalai (StartupTN official post, Global Women Summit 2026). **Loop status. Emerging.** Network and mentorship architecture exist; founder-to-founder peer financing/mentorship closure not documented; TN-RISE is barely a year old. **Assessment.** Like the SC/ST fund, an ownership-broadening instrument that has not yet closed — reinforcing the §6.3 pattern.

5.7 Niral Thiruvizha (talent funnel, pipeline documented)

Context. A mass student-innovation hackathon under Naan Mudhalvan [27]. **Provisioning/connection.**

~28,450 students from ~290 institutions; ₹10,000 prototype grants (top 1,000), ₹2 lakh vouchers (top 50), Smart Cards routing finalists into hubs [27]. **Recursion evidence.** The pipeline is documented — e.g., KCG College of Technology had two teams among the top 50 for ₹1 lakh grants (144 teams shortlisted from 837), connected to StartupTN regional hubs (KCG College official post, Sept 2024). **Loop status.** **Promising / open.** Funnel and hand-off are real; a “hackathon team became funded company” exit is not on record. **Assessment.** An excellent top-of-funnel whose recursion claim rests on conversion data the programme does not publish.

Chapter 6 — Cross-Case Analysis and Discussion

6.1 The causation problem: selector, not originator (RQ6)

Across the closed and partly-closed cases, the state’s *demonstrated* role is identification and certification, not origination. The two fully closed loops are inherited (Zoho organic; IITMRP institutional). The flagship state instrument, TANSEED, shows beneficiary relationships and scaling but no counterfactual; its showcase cases are selected-for-promise, so its headline ₹535 crore follow-on is at least as consistent with skilled selection as with catalysis. The honest reframe is therefore stronger and less flattering than “the first state-built loop is closing”: **the state is best evidenced as a credentialer, convener, and selector riding loops it does not clearly originate.** This is not a dismissal — good selection and signalling lower search costs for investors and confer legitimacy, which is real value — but it is a different and weaker claim than catalysis, and it is the claim the evidence supports. The one place a state plausibly caused a loop is Rajasthan’s procurement platform, which Tamil Nadu has not replicated.

6.2 The half-turn problem (RQ4)

Almost every confirmed state case demonstrates **grant → validation → scale → follow-on capital** and stops there. The missing turn is **founder → investor/mentor → next founder** — the turn that converts funded alumni into a self-sustaining flywheel, and the one turn the state can prompt but not provision. For the newest programmes this absence is plausibly just youth. For TANSEED it is becoming weak *data*: five years and eight editions in, with ~169 beneficiaries, there is still no documented case of a TANSEED graduate financing or formally mentoring a later cohort. This is suggestive rather than conclusive — the full flywheel historically takes ~15 years (Zoho’s did), and a founder must first reach liquidity — but at five years the silence is no longer purely premature.

6.3 Recursion is not ownership (RQ7)

Every loop measured here closes through follow-on funding, which *dilutes* founders and channels equity to external investors and acquirers. A vigorously recursing ecosystem can therefore be an ownership-concentrating one. Two observations sharpen this:

1. **The closed loop is narrow.** The only fully closed loop (Zoho/Freshworks SaaS) is urban, English-educated, and elite. Recursion here is real *and* concentrated among the already-advantaged.
2. **The broadening instruments are the open ones.** The programmes explicitly designed to widen who owns — the SC/ST fund, the women-led track, rural and tier-2 efforts — are precisely the loops that have *not* closed (§§5.5–5.6).

So the state’s distributional ambitions sit exactly where recursion is weakest, while the recursion that exists

widens nothing. A closing loop among the advantaged is not ownership-broadening. (This analysis is directional: founder/cap-table-level ownership data is largely unavailable publicly, so the claim is that recursion and ownership *must be held apart*, not that concentration is quantified.)

6.4 Policy without a platform

Procurement shows a connection that exists on paper but lacks a transactional engine. Rajasthan's contrast indicates the decisive variable is operational infrastructure (e-Bazaar) plus legal reform, not policy intent [24].

6.5 What predicts closure

Loops closed where (i) the connector had a *structural* reason to keep founders engaged (IITMRP's cap-table retention; Zoho's culture), (ii) a *transactional platform* converted eligibility into transactions (Rajasthan's e-Bazaar), and (iii) enough *time* had passed for a first generation to mature into financiers. Tamil Nadu's programmes have (i) partially (TANSEED's stake), lack (ii) in procurement, and are still accumulating (iii).

6.6 Source-reliability caveat

A large share of beneficiary-level causal claims rests on StartupTN's own posts — the funder celebrating beneficiaries it has every incentive to claim credit for. These are graded Confirmed for the *relationship* and Asserted for *causation*. Independent corroboration exists for the beneficiary status of the key cases (e.g., the Headstart cohort listing for PrimeInvestor; the Stanford cohort page for REUDE) but not for the causal link.

Chapter 7 — Conclusions and Recommendations

7.1 Conclusion (RQ5, RQ6, RQ7)

Tamil Nadu has understood that ecosystems are recursive and has engineered connection points into almost every component it funds. Recursion is genuinely evidenced — but two qualifications reshape the verdict. **On causation:** none of the closed or closing loops is clearly state-caused; the two fully closed loops are inherited, and the flagship loop's evidence is selection-consistent, so the state's demonstrated role is credentialer, convener, and selector rather than originator. **On distribution:** recursion is not ownership; the one fully closed loop is narrow and elite, while the instruments built to broaden ownership are exactly the ones still open.

The defensible verdict is therefore: **ecosystem formation in Tamil Nadu is real and accelerating, but it is neither clearly state-caused nor, as yet, ownership-widening.** Recursion is necessary but not sufficient, and closure among the already-advantaged is not the same as broad-based ownership. This is the scorecard the state would not publish — which is the only kind worth writing.

7.2 Recommendations

1. **Test causation, don't assert it.** Evaluate TANSEED against a comparison group of near-miss applicants (a regression-discontinuity around the selection threshold) to separate selection from treatment. Until then, claim credentialing, not catalysis.

2. **Close the half-turn deliberately.** Create pathways and incentives for TANSEED/SC/ST/women-led alumni to become mentors and angels for later cohorts.
3. **Give procurement a platform.** Replicate Rajasthan's e-Bazaar — a transactional catalog plus legal reform, with published outcomes [\[24\]](#).
4. **Measure ownership, not just formation.** Publish founder-retention and cap-table-distribution data, and track whether the SC/ST, women-led, rural, and tier-2 loops actually close — the test of whether recursion is widening ownership or concentrating it.
5. **Protect the institutional model.** Treat IITMRP's cap-table retention as a template for structurally retaining founders.

7.3 Falsifiable indicators of genuine, broad-based closure (3–5 years)

- A credible counterfactual evaluation showing TANSEED beneficiaries outperform comparable near-miss applicants (causation, not selection).
- Named cases of TANSEED/SC/ST/women-led graduates *investing in or mentoring* later cohorts (the half-turn).
- TN realized procurement work-order value rising from ~₹0.54 crore toward peer-state levels.
- The SC/ST and women-led loops closing — beneficiaries recirculating — i.e., broadening loops, not only the elite SaaS loop, beginning to recurse.

If these move, the ecosystem is becoming self-sustaining *and* broadening. If only the elite loop continues to recurse while the rest stay open, Tamil Nadu will remain an impressively provisioned, deliberately connected ecosystem whose recursion is narrow and whose self-sustenance is still state-credentialed.

Appendix A — Recursion Audit Matrix

Initiative	Component	Connection	Recursion evidence	Loop status	Causal grade	Verification
Zoho / Freshworks network	Talent / culture	Founder → employee → founder → investor	~59 ex-Zoho founders; alumni now angels/mentors	Closed (narrow/elite)	Organic (not state)	Confirmed [2]
IIT Madras Research Park	Anchor institution	Founder–faculty–investor links	457 startups; ~80% survival; cap-table retention	Closed (institutional)	Institutional (not StartupTN)	Confirmed [4]
TANSEED	Capital (grants)	Grant → accelerator → scale	~₹40 Cr → ~₹535 Cr follow-on; ippopay, PrimeInvestor, REUDE,	Selection vs treatment unresolved	Asserted (funder-claimed)	Relationship Confirmed; causation not established

			HARVTECH			[7]
Procurement	Market access	Govt customer → credibility	TN ~5 / ₹0.54 Cr vs Rajasthan 132 / ₹25 Cr	Open (TN); closed (Rajasthan)	Rajasthan: Established-leaning	Confirmed [24]
SC/ST Startup Fund	Capital (equity/CCD)	Funding + credibility → leadership	~₹60.8 Cr / ~26+ startups	Emerging (broadening instrument)	Not tested	Confirmed funding [22]
Women-led TANSEED / TN-RISE	Women founders	Peer / mentor networks	₹15L awards (Dukil, Aruvi Eco, etc.); council + MoUs	Emerging (broadening instrument)	Not tested	Confirmed awards [20]
Niral Thiruvizha	Talent / ideas	Hackathon → hub → startup	~28,450 students; KCG top-50 → ₹1L + hub linkage	Promising / open	Not tested	Confirmed pipeline [27]
iTNT Hub	Deep-tech	Industry-academia-founder	570+ colleges; Deep Tech Policy 2025–26	Emerging	Not tested	Confirmed activity [28]
Umanage TN	Convening	Temporary interactions	Annual summit; policy launches	Weak by design	n/a	Confirmed [29]

Reading the matrix: “Confirmed” verifies the beneficiary relationship and reported outcome; it does not verify that the programme caused the outcome. The “Causal grade” column carries that judgement separately. No state programme here reaches “Established” causation; Rajasthan’s procurement loop (external comparator) comes closest.

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Method note: claims are graded for evidence (Confirmed / Emerging) and, where causal, for causation (Established / Asserted / Not tested). Nearly all confirmed evidence demonstrates the grant → validation → scale arc; the founder → investor/mentor → next-founder closure remains strongest in the organic Zoho/IITMRP anchors and is not yet documented for the state's newer programmes. No state programme's catalytic role reaches Established causation in public records; several key causal claims rest on the funder's own communications and are graded Asserted. Two primary sources (the StartupTN women-led TANSEED awards post and the KCG Niral Thiruvizha finalists post) were reviewed directly but lacked stable public

URLs and are cited descriptively by issuer and date.